Important Note: Read Before You Start

What does this book of notes cover?

Every numbered Exercise and every Review Exercise of all 7 units of Mathematics-3 (PCTB, SNC 2020 Experimental Edition), solved step by step from the book itself. Unit 1 Whole Numbers, Unit 2 Number Operations, Unit 3 Fractions, Unit 4 Measurement (Length, Mass and Capacity), Unit 5 Measurement (Time), Unit 6 Geometry, Unit 7 Data Handling.

Are there any parts of the book that could not be solved here?

Yes, two items only, and both are flagged, not guessed. (1) Unit 3, book page 79: the Equivalent-Fractions teaching page renders near-blank in the source copy, so its teaching content cannot be read. No numbered exercise is affected. (2) Unit 4, Exercise 1, Question 7: the question text is missing from the source copy (only the 300 m by 200 m ground picture remains), so it is left unsolved. Both await a legible copy of the same edition.

What rounding rule does this book itself use for the nearest 100?

The book's own worked table (book page 30) rounds 37 to 0 to the nearest 100, because 37 is nearer to 0 than to 100. This book follows that rule throughout: for example, 46 rounds to 50 to the nearest 10 and to 0 to the nearest 100.

A note on the measurements in Unit 6.

The ruler lengths in Unit 6, Exercise 1 Question 2 and Exercise 2 Question 2 were measured from the PDF page scale. If the book is printed at a different size, the ruler readings on the printed page may differ slightly.

A note on the tomato in Unit 7, Exercise 1 Question 1.

Tomato is placed with the vegetables, following the usual school convention. Botanically a tomato is a fruit, but in everyday cooking it is treated as a vegetable.

Unit 1: Whole Numbers (book pages 1-35)

Exercise 1 (book page 5)

Q1. Write the Roman numerals for 2, 5, 12, 16, 7, 9, 19 and 20.

Step 1: Recall the symbols: I = 1, V = 5, X = 10. Step 2: Build each number: 2 = I + I = II; 5 = V; 12 = X + I + I = XII; 16 = X + V + I = XVI; 7 = V + I + I = VII; 9 = X - I = IX; 19 = X + IX = XIX; 20 = X + X = XX. Answer: 2 = II, 5 = V, 12 = XII, 16 = XVI, 7 = VII, 9 = IX, 19 = XIX, 20 = XX.

Q2. Count the chocolates in the picture and write the number in Roman numerals.

Step 1: Count the top row: 5 chocolates. Step 2: Count the bottom row: 5 chocolates. Step 3: Add: 5 + 5 = 10. Step 4: 10 in Roman numerals is X. Answer: X.

Q3. Count the dice in the pile and write the number in Roman numerals.

Step 1: Top layer: 1 die. Step 2: Middle layer: 2 dice. Step 3: Bottom layer: 3 dice. Step 4: Add: 1 + 2 + 3 = 6. Step 5: 6 in Roman numerals is VI. Answer: VI.

Q4. Fill in the missing number or Roman numeral: 6, IX, 15, XVII, XX.

Step 1: 6 = VI. Step 2: IX = 9. Step 3: 15 = XV. Step 4: XVII = 17. Step 5: XX = 20. Answer: 6 = VI, IX = 9, 15 = XV, XVII = 17, XX = 20.

Exercise 2 (book pages 9-10)

Q1. Count each group of objects and write whether the number is even or odd: blocks, sharpeners, bags, tractors, pomegranates, pizzas, peaches, pears.

Step 1: Count each group. Step 2: A number is even if its ones digit is 0, 2, 4, 6 or 8; it is odd if its ones digit is 1, 3, 5, 7 or 9. Answer: blocks 3 Odd (given in the book); sharpeners 10 Even; bags 4 Even; tractors 2 Even; pomegranates 5 Odd; pizzas 5 Odd; peaches 9 Odd; pears 8 Even.

Q2. From the grid (1, 8, 7, 10, 23, 14, 15, 30, 35, 42, 55, 65, 68, 72, 79, 82, 83, 91, 96, 99), separate the even and odd numbers.

Step 1: Check the ones digit of each number. Step 2: Even ones digits are 0, 2, 4, 6, 8. Step 3: Odd ones digits are 1, 3, 5, 7, 9. Answer: Even: 8, 10, 14, 30, 42, 68, 72, 82, 96. Odd: 1, 7, 23, 15, 35, 55, 65, 79, 83, 91, 99.

Q3. Write the odd numbers between (i) 4 and 16, (ii) 20 and 34.

Step 1: Odd numbers have ones digit 1, 3, 5, 7 or 9. Step 2: List them in order, without the end numbers. Answer: (i) Between 4 and 16: 5, 7, 9, 11, 13, 15. (ii) Between 20 and 34: 21, 23, 25, 27, 29, 31, 33.

Q4. Write the even numbers between (i) 1 and 10, (ii) 21 and 35.

Step 1: Even numbers have ones digit 0, 2, 4, 6 or 8. Step 2: List them in order, without the end numbers. Answer: (i) Between 1 and 10: 2, 4, 6, 8. (ii) Between 21 and 35: 22, 24, 26, 28, 30, 32, 34.

Q5. From (2, 5, 9, 18, 21, 28, 35, 56, 67, 73, 79, 80, 84, 87, 90, 93, 94, 95, 97, 99), separate the even and odd numbers.

Step 1: Check the ones digit of each number. Answer: Even: 2, 18, 28, 56, 80, 84, 90, 94. Odd: 5, 9, 21, 35, 67, 73, 79, 87, 93, 95, 97, 99.

Exercise 3 (book pages 14-17)

Q1. Write in words: 5342, 7123, 5321, 8035, 9899, 80321.

Step 1: Read each place value from left to right. Answer: 5342 = Five thousand three hundred forty-two; 7123 = Seven thousand one hundred twenty-three; 5321 = Five thousand three hundred twenty-one; 8035 = Eight thousand thirty-five; 9899 = Nine thousand eight hundred ninety-nine; 80321 = Eighty thousand three hundred twenty-one.

Q2. Write in numerals: Five thousand eight hundred forty; Six thousand three hundred sixty-three; Thirty-two thousand three hundred eight; Eighty thousand five hundred eighty-seven; Sixty-four thousand thirty-three; Forty-one thousand nine hundred ninety-nine.

Step 1: Write the thousands part, then the hundreds, tens and ones. Answer: 5840; 6363; 32308; 80587; 64033; 41999.

Q3. Write the expanded form: 2347, 6780, 34560, 53406, 92341.

Step 1: Write each digit with its place (Ten-Thousands, Thousands, Hundreds, Tens, Ones). Answer: 2347 = 2 Th + 3 H + 4 T + 7 O; 6780 = 6 Th + 7 H + 8 T + 0 O; 34560 = 3 Ten-Th + 4 Th + 5 H + 6 T + 0 O; 53406 = 5 Ten-Th + 3 Th + 4 H + 0 T + 6 O; 92341 = 9 Ten-Th + 2 Th + 3 H + 4 T + 1 O.

Q4. Match the words with the numerals: Seven thousand eight hundred; Eighty-two thousand six hundred fifty-one; Fifteen thousand seven hundred sixty-three; Fifty-three thousand one hundred two; Two thousand three hundred fifty-seven; Five thousand three hundred twenty-six; Three hundred eighty-four.

Step 1: Convert each word form to numerals. Answer: Seven thousand eight hundred = 7800; Eighty-two thousand six hundred fifty-one = 82651; Fifteen thousand seven hundred sixty-three = 15763; Fifty-three thousand one hundred two = 53102; Two thousand three hundred fifty-seven = 2357; Five thousand three hundred twenty-six = 5326; Three hundred eighty-four = 384.

Q5. In 4869, what is the value of 4 and what is the value of 6?

Step 1: In 4869, 4 is in the thousands place, so its value is 4 x 1000 = 4000.

Step 2: 6 is in the tens place, so its value is 6 x 10 = 60. Answer: Value of 4 = 4000 (thousands); value of 6 = 60 (tens).

Q6. Write the place value of each digit: 2357, 67815, 82301, 75389.

Step 1: Multiply each digit by its place value. Answer: 2357: 2000, 300, 50, 7. 67815: 60000, 7000, 800, 10, 5. 82301: 80000, 2000, 300, 0, 1. 75389: 70000, 5000, 300, 80, 9.

Q7. Write the place value of the encircled digit: (i) 4567 (6 encircled), (ii) 5327 (3 encircled), (iii) 85761 (5 encircled), (iv) 70431 (1 encircled), (v) 67431 (6 encircled), (vi) 39761 (7 encircled), (vii) 93267 (6 encircled), (viii) 68037 (6 encircled), (ix) 54136 (4 encircled), (x) 89791 (9 encircled).

Step 1: Find the place of the encircled digit and multiply. Answer: (i) 60; (ii) 300; (iii) 5000; (iv) 1; (v) 60000; (vi) 700; (vii) 60; (viii) 60000; (ix) 4000; (x) 900.

Q8. (i) Five children hold the digit cards 7, 3, 8, 5, 2 in order. What number do they make, and what is the place value of each digit? (ii) A train has compartments numbered 4, 6, 3, 1, 9 in order. What number do they make, and what is the place value of each digit?

Step 1: Join the digits in order. Step 2: Write each digit's place value. Answer: (i) Number = 73852. Place values: 70000, 3000, 800, 50, 2. (ii) Number = 46319. Place values: 40000, 6000, 300, 10, 9.

Crossword activity (book pages 17-18)

Crossword numbers: (i) 7343 is already done (vertical, column 3, rows 3-6). (ii) 37729 goes in row 3, columns 6-10. (iii) 6371 goes in column 6, rows 2-5. (iv) 4598 goes in row 4, columns 8-10. (v) 72271 goes in column 5, rows 2-6. (vi) 35276 goes in row 4, columns 3-7.

Crossword: convert and fill: (i) Seven thousand three hundred forty three; (ii) Thirty seven thousand seven hundred twenty nine; (iii) Six thousand three hundred seventy one; (iv) 4 Th + 5 H + 9 T + 8 O; (v) 7 Ten-Th + 2 Th + 2 H + 7 T + 1 O; (vi) 3 Ten-Th + 5 Th + 2 H + 7 T + 6 O.

Step 1: Convert each form to numerals. Answer: (i) 7343; (ii) 37729; (iii) 6371; (iv) 4598; (v) 72271; (vi) 35276.

Make three smaller numbers from 7152 (examples shown in the book): 1752, 1572, 5127. Any three different numbers made from the digits 7, 1, 5, 2 that are smaller than 7152 are correct.

Exercise 4 (book page 21)

Q1. Mark the given numbers on the number line: (i) 9, 4; (ii) 5, 15, 25; (iii) 6, 12, 24, 36; (iv) 4, 12, 20, 28; (v) 0, 15, 30, 60; (vi) 3, 6, 9, 12, 15.

Step 1: Find the step of each number line. Step 2: Count forward by that step and mark each number. Answer: (i) Mark 9 and 4. (ii) Mark 5, 15 and 25. (iii) Mark 6, 12, 24 and 36. (iv) Mark 4, 12, 20 and 28. (v) Mark 0, 15, 30 and 60. (vi) Mark 3, 6, 9, 12 and 15.

Q2. Find the value of each picture on the number line: (i) step 5, a bag and an apple; (ii) step 10, a yellow box, a red box and a pink box; (iii) step 2, a tractor, a sharpener, books and dice.

Step 1: Count the steps from 0 to each picture. Answer: (i) Step 5: bag = 10, apple = 20. (ii) Step 10: yellow box = 30, red box = 50, pink box = 70. (iii) Step 2: tractor = 6, sharpener = 10, books = 14, dice = 16.

Q3. Place the number cards in the empty boxes on the number line: (a) line 0-24, step 3, cards 21, 18, 9, 6; (b) line 0-800, step 100, cards 200, 700, 500.

Step 1: Write the number line marks. Step 2: Put each card on its matching mark, left to right. Answer: (a) Boxes left to right: 6, 9, 18, 21. (b) Boxes left to right: 200, 500, 700.

Exercise 5 (book pages 27-28)

Q1. Put the correct symbol (>, < or =): (i) 873 and 426; (ii) 694 and 706; (iii) 857 and 857; (iv) 978 and 824; (v) 574 and 574; (vi) 819 and 817.

Step 1: Compare the hundreds digits first, then tens, then ones. Answer: (i) 873 > 426; (ii) 694 < 706; (iii) 857 = 857; (iv) 978 > 824; (v) 574 = 574; (vi) 819 > 817.

Q2. Write the greater number in each pair (pairs as given in the book).

Step 1: Compare hundreds, then tens, then ones. Answer: 671; 249; 374; 738; 937; 877.

Q3. Write in ascending and descending order: (i) 71, 51, 91, 61; (ii) 85, 52, 73, 41, 67; (iii) 346, 451, 321, 536; (iv) 698, 278, 543, 231, 731; (v) 476, 471, 472, 335, 345.

Step 1: Ascending means smallest to greatest. Step 2: Descending means greatest to smallest. Answer: (i) Asc: 51, 61, 71, 91. Desc: 91, 71, 61, 51. (ii) Asc: 41, 52, 67, 73, 85. Desc: 85, 73, 67, 52, 41. (iii) Asc: 321, 346, 451, 536. Desc: 536, 451, 346, 321. (iv) Asc: 231, 278, 543, 698, 731. Desc: 731, 698, 543, 278, 231. (v) Asc: 335, 345, 471, 472, 476. Desc: 476, 472, 471, 345, 335.

Q4. Write five numbers less than 321, then write them in ascending and descending order.

Step 1: Any five numbers below 321 are correct; the book's example uses 301, 305, 310, 315, 320. Step 2: Order them. Answer: Asc: 301, 305, 310, 315, 320. Desc: 320, 315, 310, 305, 301.

Exercise 6 (book pages 31-32)

Q1. Round to the nearest 10 and to the nearest 100: 26, 52, 327, 385, 750.

Step 1: For the nearest 10, look at the ones digit: 5 or more rounds up. Step 2: For the nearest 100, look at the tens digit: 5 or more rounds up; remember the book's rule that a number nearer to 0 rounds to 0. Answer: 26 to 30, 0. 52 to 50, 100. 327 to 330, 300. 385 to 390, 400. 750 to 750, 800.

Q2. Match each number with its nearest 10 and nearest 100: nearest 10: 19, 32, 41, 52, 61, 76, 95; nearest 100: 101, 256, 384, 517, 649, 739, 850.

Step 1: Round each number. Answer: Nearest 10: 19 to 20, 32 to 30, 41 to 40, 52 to 50, 61 to 60, 76 to 80, 95 to 100. Nearest 100: 101 to 100, 256 to 300, 384 to 400, 517 to 500, 649 to 600, 739 to 700, 850 to 900.

Q3. On the number line, 26 rounds to which nearest 10?

Step 1: On the number line, 26 lies between 20 and 30. Step 2: 26 is nearer to 30. Answer: 30.

Q4. On the number line, 735 rounds to which nearest 100?

Step 1: On the number line, 735 lies between 700 and 800. Step 2: 735 is nearer to 700. Answer: 700.

Review Exercise (book pages 33-35)

Q1 (i). Roman number XIX is equal to:

(a) 10

(b) 11

✓ (c) 19

(d) 20

Q1 (ii). Place value of the digit 2 in 2750 is:

(a) tens

(b) ten thousands

✓ (c) thousands

(d) hundreds

Q1 (iii). Eight thousand seven hundred twenty is:

✓ (a) 8720

(b) 8702

(c) 8072

(d) 87020

Q1 (iv). 23, 25, 21 and 27 can be written in descending order as:

(a) 21, 23, 25, 27

(b) 23, 25, 27

(c) 27, 23, 21

✓ (d) 27, 25, 23, 21

Q1 (v). 16 can be rounded off to the nearest 10 as:

(a) 10

(b) 15

(c) 20

(d) 16

25 can be rounded off as 30 to the nearest 10.

In ascending order, numbers are written from lowest to highest.

Number of wheels in a car are 4, which is even.

Number of sides of a triangle are 3, which is odd.

In an odd number, the digits at ones place are 1, 3, 5, 7, 9.

Q3. Write the Roman numerals for 2, 5, 8, 11, 15.

Step 1: Use I = 1, V = 5, X = 10. Answer: 2 = II, 5 = V, 8 = VIII, 11 = XI, 15 = XV.

Q4. Write the place value of the encircled digit: 5342 (3 encircled), 7063 (0 encircled), 12865 (5 encircled), 80064 (0 encircled), 96563 (6 encircled).

Step 1: Find the place of the encircled digit. Answer: 300; 0; 5; 0; 60.

Q5. Write in words: 647, 7265, 9999, 9765, 8701.

Answer: 647 = six hundred forty-seven; 7265 = seven thousand two hundred sixty-five; 9999 = nine thousand nine hundred ninety-nine; 9765 = nine thousand seven hundred sixty-five; 8701 = eight thousand seven hundred one.

Q6. Write in ascending and descending order: (i) 27, 21, 3, 45; (ii) 512, 321, 445, 241, 114.

Answer: (i) Asc: 3, 21, 27, 45. Desc: 45, 27, 21, 3. (ii) Asc: 114, 241, 321, 445, 512. Desc: 512, 445, 321, 241, 114.

Q7. From the numbers in the book, separate the even and odd numbers.

Step 1: Check the ones digit of each number. Answer: Even: 34, 64, 84, 88. Odd: 15, 45, 71, 77.

Q8. Complete the number line with step 50, from 0 to 450.

Step 1: Start at 0 and add 50 each time. Answer: 0, 50, 100, 150, 200, 250, 300, 350, 400, 450.

Q9. Put the correct symbol between 928 and 985.

Step 1: Compare hundreds: both 9. Step 2: Compare tens: 2 is less than 8. Answer: 928 < 985.

Q10. Round to the nearest 10 and to the nearest 100: 46, 83, 765, 847, 956.

Step 1: Round to the nearest 10 by the ones digit; to the nearest 100 by the tens digit, using the book's rule (46 is nearer to 0 than to 100). Answer: 46 to 50, 0. 83 to 80, 100. 765 to 770, 800. 847 to 850, 800. 956 to 960, 1000.

Review activity (book page 35): Count the children in Suleman's birthday picture. There are 4 girls and 6 boys. Both groups have an even number of people.

Unit 2: Number Operations (book pages 36-67)

Exercise 1 (book pages 40-42)

Q1. Add (without carrying where possible): 6643 + 3215; 5137 + 3542; 7256 + 1423; 6795 + 2104; 7000 + 2137; 4000 + 3154.

Step 1: Add ones, then tens, then hundreds, then thousands. Answer: 6643 + 3215 = 9858 (ones 3+5=8, tens 4+1=5, hundreds 6+2=8, thousands 6+3=9). 5137 + 3542 = 8679. 7256 + 1423 = 8679. 6795 + 2104 = 8899. 7000 + 2137 = 9137. 4000 + 3154 = 7154.

Q2. Add (with carrying): 5794 + 3825; 4752 + 3596; 5496 + 2179; 6243 + 5727; 6495 + 2156; 5676 + 3864.

Step 1: Add ones; carry the ten if the sum is 10 or more. Step 2: Add tens with the carry, then hundreds, then thousands. Answer: 5794 + 3825 = 9619. 4752 + 3596 = 8348. 5496 + 2179 = 7675. 6243 + 5727 = 11970. 6495 + 2156 = 8651. 5676 + 3864 = 9540.

Q3. There are 3454 trees in one park and 2345 trees in another park. How many trees are there in all?

Step 1: Add the two numbers: 3454 + 2345. Step 2: Ones 4+5=9, tens 5+4=9, hundreds 4+3=7, thousands 3+2=5. Answer: 5799 trees.

Q4. Add: 6758 + 3441 (Rs).

Step 1: Add: 6758 + 3441. Step 2: Ones 8+1=9, tens 5+4=9, hundreds 7+4=11 (write 1, carry 1), thousands 6+3+1=10. Answer: Rs 10199.

Q5. 4536 people visit a fair on Saturday and 3253 people visit on Sunday. How many people visit in all?

Step 1: Add: 4536 + 3253 = 7789. Answer: 7789 people.

Q6. A factory has 6540 employees in one unit and 2120 employees in another unit. How many employees are there in all?

Step 1: Add: 6540 + 2120 = 8660. Answer: 8660 employees.

Q7. Ali and Shahnawaz save Rs 4056 and Rs 5430. How much do they save together?

Step 1: Add: 4056 + 5430. Step 2: Ones 6+0=6, tens 5+3=8, hundreds 0+4=4, thousands 4+5=9. Answer: Rs 9486.

Q8. Add mentally: 15 + 16; 52 + 18; 59 + 10; 47 + 32; 35 + 55; 46 + 24; 37 + 23; 36 + 54; 27 + 43; 56 + 24; 42 + 15; 32 + 28.

Step 1: Add tens first, then ones. Answer: 31; 70; 69; 79; 90; 70; 60; 90; 70; 80; 57; 60.

Exercise 2 (book pages 47-48)

Q1. Subtract: 3546 - 2324; 5796 - 3453; 6354 - 4041; 8764 - 3653; 4754 - 3532; 9876 - 6754; 9765 - 8974; 8754 - 3974; 6495 - 3546; 7965 - 6876; 8678 - 7896; 8543 - 7654.

Step 1: Subtract ones, then tens, then hundreds, then thousands, borrowing where the top digit is smaller. Answer: 1222; 2343; 2313; 5111; 1222; 3122; 791; 4780; 2949; 1089; 782; 889.

Q2. Subtract mentally: 80 - 24; 65 - 41; 67 - 25; 76 - 35; 87 - 36; 57 - 21.

Step 1: Subtract tens first, then ones. Answer: 56; 24; 42; 41; 51; 36.

Q3. A book has 1535 pages. 424 pages have been read. How many pages are left to read?

Step 1: Subtract: 1535 - 424. Step 2: Ones 5-4=1, tens 3-2=1, hundreds 5-4=1, thousands 1-0=1. Answer: 1111 pages left.

Q4. Subtract: 6456 - 4340 (Rs).

Step 1: Subtract: 6456 - 4340 = 2116. Answer: Rs 2116.

Q5. There are 6753 people in a town. 3985 are women. How many are men?

Step 1: Subtract: 6753 - 3985. Step 2: Borrow where needed: ones 13-5=8, tens 14-8=6, hundreds 16-9=7, thousands 6-1-3=2. Answer: 2768 men.

Q6. A farmer has 7516 goats. He sells 5728 goats. How many goats are left?

Step 1: Subtract: 7516 - 5728 = 1788. Answer: 1788 goats.

Exercise 3 (book page 53)

Q1. Write the multiplication tables of 6, 7, 8 and 9 (1 to 10).

Step 1: Multiply each number by 1, 2, 3, up to 10. Answer: 6 times: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60. 7 times: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. 8 times: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80. 9 times: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90.

Q2. Complete the multiplication grid for 6, 7, 8 and 9.

Step 1: Fill each row by multiplying. Given in the book: 6 x 1 = 6, 6 x 5 = 30, 6 x 9 = 54; 7 x 2 = 14, 7 x 6 = 42, 7 x 10 = 70; 8 x 3 = 24, 8 x 7 = 56; 9 x 4 = 36, 9 x 8 = 72. Answer: 6 row: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60. 7 row: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. 8 row: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80. 9 row: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90.

Exercise 4 (book pages 59-61)

Q1. Multiply: 24 x 3; 35 x 4; 32 x 5; 38 x 6; 45 x 7; 48 x 8; 54 x 9; 56 x 7; 62 x 6.

Step 1: Multiply the ones digit first, carry the ten, then multiply the tens digit and add the carry. Answer: 24 x 3 = 72. 35 x 4 = 140. 32 x 5 = 160. 38 x 6 = 228. 45 x 7 = 315. 48 x 8 = 384. 54 x 9 = 486. 56 x 7 = 392. 62 x 6 = 372.

Q2. Multiply using the tables: 7 x 6; 5 x 6; 4 x 7; 9 x 7; 4 x 9; 8 x 7.

Answer: 7 x 6 = 42. 5 x 6 = 30. 4 x 7 = 28. 9 x 7 = 63. 4 x 9 = 36. 8 x 7 = 56.

5 x 0 = 0.

35 x 0 = 0.

45 x 0 = 0.

48 x 1 = 48.

1 x 57 = 57.

31 x 1 = 31.

Q4. Multiply mentally: 4 x 9; 5 x 4; 6 x 3; 6 x 5; 8 x 4; 7 x 5.

Answer: 36; 20; 18; 30; 32; 35.

Q5. Multiply: 24 x 4 (Rs).

Step 1: Multiply the price by the number: 24 x 4. Step 2: Ones 4 x 4 = 16 (write 6, carry 1). Tens 2 x 4 = 8, plus carry 1 = 9. Answer: Rs 96.

Q6. There are 52 weeks in a year. How many weeks are there in 7 years?

Step 1: Multiply: 52 x 7. Step 2: Ones 2 x 7 = 14 (write 4, carry 1). Tens 5 x 7 = 35, plus carry 1 = 36. Answer: 364 weeks.

Q7. There are 28 trees in one row. How many trees are there in 5 such rows?

Step 1: Multiply: 28 x 5. Step 2: Ones 8 x 5 = 40 (write 0, carry 4). Tens 2 x 5 = 10, plus carry 4 = 14. Answer: 140 trees.

Q8. A car travels 62 km in 1 litre of petrol. How far will it travel in 4 litres?

Step 1: Multiply: 62 x 4. Step 2: Ones 2 x 4 = 8. Tens 6 x 4 = 24. Answer: 248 km.

Exercise 5 (book pages 64-65)

Divide: 20 / 4; 25 / 5; 42 / 6; 49 / 7; 72 / 8; 81 / 9; 48 / 4; 72 / 6; 84 / 7; 51 / 3; 96 / 8; 99 / 9; 4 / 2; 8 / 4; 9 / 3.

Step 1: Use the multiplication tables in reverse. Answer: 20 / 4 = 5. 25 / 5 = 5. 42 / 6 = 7. 49 / 7 = 7. 72 / 8 = 9. 81 / 9 = 9. 48 / 4 = 12. 72 / 6 = 12. 84 / 7 = 12. 51 / 3 = 17. 96 / 8 = 12. 99 / 9 = 11. 4 / 2 = 2. 8 / 4 = 2. 9 / 3 = 3.

Q16. 96 students stand in 6 equal rows. How many students are in each row?

Step 1: Divide: 96 / 6. Step 2: 6 x 16 = 96. Answer: 16 students in each row.

Q17. A man covers 56 km in 4 days, the same distance each day. How many km does he cover in one day?

Step 1: Divide: 56 / 4 = 14. Answer: 14 km per day.

Q18. 70 pencils are packed equally in 5 packets. How many pencils are in each packet?

Step 1: Divide: 70 / 5 = 14. Answer: 14 pencils in each packet.

Q19. 91 rupees buy 7 notebooks of equal price. What is the price of one notebook?

Step 1: Divide: 91 / 7 = 13. Answer: Rs 13 per notebook.

Q20. 48 pencils are shared equally among 8 children. How many pencils does each child get?

Step 1: Divide: 48 / 8 = 6. Answer: 6 pencils.

Review Exercise (book pages 66-67)

Q1 (i). Sum of 1564 and 7325 is:

(a) 8888

✓ (b) 8889

(c) 8899

(d) 8886

Q1 (ii). Difference of 6351 and 1265 is:

(a) 5056

(b) 5076

✓ (c) 5086

(d) 5096

Q1 (iii). 3246 is _____ less than 4596:

✓ (a) 1350

(b) 1360

(c) 1370

(d) 1380

Q1 (iv). There are 6 eggs in a basket. Then _____ eggs are in 7 such baskets:

(a) 21

(b) 28

(c) 35

✓ (d) 42

Q1 (v). When any number is multiplied by 0, we get:

✓ (a) 0

(b) 1

(c) 10

(d) 100

Q1 (vi). By multiplying 12 by 1, we get:

(a) 13

(b) 112

✓ (c) 12

(d) 14

Q1 (vii). By dividing 24 by 6, we get:

✓ (a) 4

(b) 5

(c) 6

(d) 7

Q1 (viii). By dividing 84 by 4, we get:

(a) 18

(b) 19

(c) 20

✓ (d) 21

Q2. Add: 4536 + 5314.

Step 1: Ones 6+4=10 (write 0, carry 1). Tens 3+1+1=5. Hundreds 5+3=8.

Thousands 4+5=9. Answer: 9850.

Q3. Add: 8645 + 3456.

Step 1: Ones 5+6=11 (write 1, carry 1). Tens 4+5+1=10 (write 0, carry 1). Hundreds 6+4+1=11 (write 1, carry 1). Thousands 8+3+1=12. Answer: 12101.

Q4. Subtract: 4554 - 2342.

Step 1: Ones 4-2=2, tens 5-4=1, hundreds 5-3=2, thousands 4-2=2. Answer: 2212.

Q5. Subtract: 5943 - 4864.

Step 1: Borrow where needed: ones 13-4=9, tens 13-6=7, hundreds 8-8=0, thousands 5-4=1. Answer: 1079.

Q6. Add: 28 + 13.

Step 1: Ones 8+3=11 (write 1, carry 1). Tens 2+1+1=4. Answer: 41.

Q7. Subtract: 58 - 32.

Step 1: Ones 8-2=6, tens 5-3=2. Answer: 26.

Q8. Multiply: 8 x 6.

Step 1: 8 times 6 = 48. Answer: 48.

Q9. Divide: 6 / 3.

Step 1: 3 x 2 = 6. Answer: 2.

Q10. Add: 1457 + 7321 (fruits).

Step 1: Add: 1457 + 7321 = 8778. Answer: 8778 fruits.

Q11. A school has 4356 students. 1968 are female students. How many are male students?

Step 1: Subtract: 4356 - 1968 = 2388. Answer: 2388 male students.

Q12. Multiply: 35 x 7 (Rs).

Step 1: Multiply: 35 x 7 = 245. Answer: Rs 245.

Q13. 8 kg cost Rs 96. What is the price of 1 kg?

Step 1: Divide: 96 / 8 = 12. Answer: Rs 12 per kg.

Unit 3: Fractions (book pages 68-87)

Exercise 1 (book pages 70-72)

Q1. Write the numerator and denominator of each fraction: 2/7, 3/7, 5/8, 2/5, 10/13, 9/10, 1/8, 2/3, 4/7, 3/4.

Step 1: The top number is the numerator; the bottom number is the denominator. Answer: 2/7 (numerator 2, denominator 7); 3/7 (3, 7); 5/8 (5, 8); 2/5 (2, 5); 10/13 (10, 13); 9/10 (9, 10); 1/8 (1, 8); 2/3 (2, 3); 4/7 (4, 7); 3/4 (3, 4).

Q2. Write the fraction of the coloured part: first circle 1/4 (given in the book), second circle, third circle, first grid, second grid, square of 8 triangles.

Step 1: Count the total equal parts (denominator). Step 2: Count the coloured parts (numerator). Answer: Circles: 1/4 (given), 3/8, 1/3. Grids: 5/16, 4/16. Square of 8 triangles: 5/8 (5 green triangles out of 8).

Q3. Colour the given fraction of each figure: 2/3, 3/4, 3/8, 3/6.

Step 1: Divide the figure into the denominator's equal parts. Step 2: Colour the numerator's number of parts. Answer: Colour 2 of 3 parts for 2/3; 3 of 4 triangles for 3/4; 3 of 8 sectors for 3/8; 3 of 6 parts for 3/6.

Q4. Write the fraction shown by each figure: (i) 4/11 (given in the book), (ii), (iii), (iv).

Step 1: Count the total parts and the shaded parts. Answer: (i) 4/11 (given); (ii) 3/11; (iii) 4/9; (iv) 5/7.

Q5. Match each figure with its fraction: (i) 2/3, (ii) 1/4, (iii) 2/5, (iv) 3/8, (v) 2/8, (vi) 3/4.

Step 1: Count the shaded parts and total parts of each figure. Answer: (i) 2/3; (ii) 1/4; (iii) 2/5; (iv) 3/8; (v) 2/8; (vi) 3/4.

Exercise 2 (book pages 74-75)

Q1. Tell whether each fraction is proper or improper: 3/4, 4/5, 4/3, 4/9, 7/5, 9/5, 8/9, 3/7, 7/7.

Step 1: If the numerator is smaller than the denominator, the fraction is proper.

Step 2: If the numerator is greater than or equal to the denominator, it is improper (the book's Key Fact). Answer: (i) 3/4 proper; (ii) 4/5 proper; (iii) 4/3 improper; (iv) 4/9 proper; (v) 7/5 improper; (vi) 9/5 improper; (vii) 8/9 proper; (viii) 3/7 proper; (ix) 7/7 improper.

Q2. Separate into proper and improper fractions: 4/5, 13/6, 7/12, 14/5, 7/19, 9/4, 8/15, 8/5, 7/9, 7/4.

Step 1: Compare numerator and denominator of each fraction. Answer: Proper: 4/5, 7/12, 7/19, 8/15, 7/9. Improper: 13/6, 14/5, 9/4, 8/5, 7/4.

Exercise 3 (book page 78)

Q1. Match each fraction with its equivalent fraction: 3/5, 5/9, 4/7, 3/6, 3/8, 5/7.

Step 1: Multiply numerator and denominator by the same number. Answer: 3/5 = 6/10; 5/9 = 10/18; 4/7 = 8/14; 3/6 = 1/2; 3/8 = 9/24; 5/7 = 15/21.

Q2. Write three equivalent fractions for each: 5/6, 2/3, 1/4, 5/8, 3/5, 2/5.

Step 1: Multiply numerator and denominator by 2, then by 3, then by 4. Answer: (i) 5/6: 10/12, 15/18, 20/24. (ii) 2/3: 4/6, 6/9, 8/12. (iii) 1/4: 2/8, 3/12, 4/16. (iv) 5/8: 10/16, 15/24, 20/32. (v) 3/5: 6/10, 9/15, 12/20. (vi) 2/5: 4/10, 6/15, 8/20.

Note on book page 79: The Equivalent-Fractions teaching page on book page 79 renders near-blank in the source copy (only a yellow brush, empty bubbles and part of a bread picture can be seen). Its teaching content cannot be read here and is not guessed. No numbered exercise is affected: Exercise 3 ends on book page 78 and Exercise 4 starts on book page 80.

Exercise 4 (book page 80)

Q1. Put the correct symbol (>, < or =): (i) 5/8 and 7/8; (ii) 3/4 and 2/4; (iii) 1/3 and 2/3; (iv) 5/8 and 3/8.

Step 1: With the same denominator, the fraction with the greater numerator is greater. Answer: (i) 5/8 < 7/8; (ii) 3/4 > 2/4; (iii) 1/3 < 2/3; (iv) 5/8 > 3/8.

Q2. Put the correct symbol: (i) 3/9 and 5/9; (ii) 3/5 and 2/5; (iii) 4/7 and 4/7; (iv) 2/3 and 1/3; (v) 4/9 and 4/9; (vi) 5/11 and 3/11.

Answer: (i) 3/9 < 5/9; (ii) 3/5 > 2/5; (iii) 4/7 = 4/7; (iv) 2/3 > 1/3; (v) 4/9 = 4/9; (vi) 5/11 > 3/11.

Exercise 5 (book page 82)

Add (same denominators): 3/7 + 2/7; 3/5 + 1/5; 1/9 + 4/9; 5/12 + 2/12; 1/8 + 3/8; 1/6 + 3/6; 1/3 + 1/3; 3/10 + 4/10; 3/8 + 2/8.

Step 1: Keep the denominator the same. Step 2: Add the numerators. Answer: 5/7; 4/5; 5/9; 7/12; 4/8; 4/6; 2/3; 7/10; 5/8.

Exercise 6 (book page 84)

Subtract (same denominators): 3/7 - 1/7; 5/9 - 1/9; 3/5 - 2/5; 5/8 - 2/8; 7/12 - 3/12; 5/6 - 3/6; 5/8 - 3/8; 5/11 - 3/11; 7/15 - 3/15; 3/4 - 1/4; 5/8 - 2/8; 7/10 - 3/10.

Step 1: Keep the denominator the same. Step 2: Subtract the numerators.

Answer: 2/7; 4/9; 1/5; 3/8; 4/12; 2/6; 2/8; 2/11; 4/15; 2/4; 3/8; 4/10.

Review Exercise (book pages 85-87)

Q1 (i). A fraction in which numerator is smaller than denominator is called _____ fraction:

- ✓ (a) proper
- (b) improper
- (c) equivalent
- (d) common

Q1 (ii). A fraction in which numerator is greater than denominator is called _____ fraction:

(a) equivalent

(b) common

(c) proper

✓ (d) improper

Q1 (iii). Equivalent fraction of 2/5 is:

(a) 4/3

(b) 4/7

(c) 4/6

✓ (d) 4/10

Q1 (iv). In fractions, 4/5 [ ] 3/5, use symbol:

(a) <

✓ (b) >

(c) =

(d) not equal

Q1 (v). In fractions, 4/15 [ ] 7/15, use symbol:

(a) >

✓ (b) <

(c) =

(d) not equal

Q1 (vi). The sum of two fractions 3/15 and 4/15 is:

(a) 1/15

✓ (b) 7/15

(c) 7/30

(d) 1/30

Q1 (vii). The difference of two fractions 7/9 and 3/9 is:

(a) 4/9

(b) 10/9

(c) 10/18

(d) 4/18

Q2. Identify numerators and denominators of: 2/9, 3/7, 4/5, 10/7, 4/15, 11/6.

Answer: 2/9 (numerator 2, denominator 9); 3/7 (3, 7); 4/5 (4, 5); 10/7 (10, 7); 4/15 (4, 15); 11/6 (11, 6).

Q3. Separate proper and improper fractions from: 3/5, 7/5, 9/6, 3/8, 5/9, 6/6, 7/18.

Step 1: Numerator smaller than denominator means proper; greater than or equal means improper. Answer: Proper: 3/5, 3/8, 5/9, 7/18. Improper: 7/5, 9/6, 6/6.

Q4. Compare using <, > or =: (i) 8/9 and 4/9; (ii) 5/7 and 6/7; (iii) 4/5 and 4/5.

Answer: (i) 8/9 > 4/9; (ii) 5/7 < 6/7; (iii) 4/5 = 4/5.

Q5. Write three equivalent fractions for: 2/3, 4/5, 3/7, 3/8.

Answer: (i) 2/3: 4/6, 6/9, 8/12. (ii) 4/5: 8/10, 12/15, 16/20. (iii) 3/7: 6/14, 9/21, 12/28. (iv) 3/8: 6/16, 9/24, 12/32.

Q6. Add: (i) 4/8 + 3/8; (ii) 3/10 + 5/10; (iii) 1/4 + 2/4.

Answer: (i) 7/8; (ii) 8/10; (iii) 3/4.

Q7. Subtract: (i) 2/3 - 1/3; (ii) 3/4 - 2/4.

Answer: (i) 1/3; (ii) 1/4.

Unit 4: Measurement - Length, Mass and Capacity (book pages 88-117)

Exercise 1 (book pages 93-94)

Q1-Q6. Add the lengths: (1) 4 m 65 cm + 5 m 12 cm; (2) 14 m 50 cm + 9 m 40 cm; (3) 7 km 632 m + 8 km 214 m; (4) 25 km 312 m + 21 km 676 m; (5) 21 km 815 m + 17 km 183 m; (6) 41 km 745 m + 38 km 134 m.

Step 1: Add the smaller unit first (cm with cm, m with m). Step 2: Add the larger unit (m with m, km with km). Answer: (1) 9 m 77 cm (65+12=77, 4+5=9). (2) 23 m 90 cm (50+40=90, 14+9=23). (3) 15 km 846 m (632+214=846, 7+8=15). (4) 46 km 988 m (312+676=988, 25+21=46). (5) 38 km 998 m (815+183=998, 21+17=38). (6) 79 km 879 m (745+134=879, 41+38=79).

Q7. (Book pages 93-94) - NOT SOLVED: question text missing from the source copy.

The question text for Question 7 is missing from the source copy of the book: only the question number 7 and a picture of a ground marked 300 m by 200 m (with a long-jump pit) remain, and the text area is blank. Because the question itself is unknown, no answer is guessed here. This question awaits a legible copy of the same edition of the book.

Exercise 2 (book pages 97-98)

Q1. Subtract the lengths: (i) 51 m 86 cm - 30 m 75 cm; (ii) 25 m 93 cm - 14 m 23 cm; (iii) 15 km 365 m - 13 km 252 m; (iv) 76 m 67 cm - 35 m 41 cm; (v) 67 km 891 m - 51 km 760 m; (vi) 35 km 786 m - 13 km 675 m; (vii) 19 km 345 m - 16 km 231 m; (viii) 19 km 708 m - 15 km 205 m.

Step 1: Subtract the smaller unit first, then the larger unit, borrowing where needed. Answer: (i) 21 m 11 cm; (ii) 11 m 70 cm; (iii) 2 km 113 m; (iv) 41 m 26 cm; (v) 16 km 131 m; (vi) 22 km 111 m; (vii) 3 km 114 m; (viii) 4 km 503 m.

Q2. A pipe is 78 m 89 cm long. 35 m 65 cm of it is cut off. How much pipe is left?

Step 1: Subtract: 78 m 89 cm - 35 m 65 cm. Step 2: cm: 89-65=24. m: 78-35=43. Answer: 43 m 24 cm pipe left.

Q3. A lace is 19 m 82 cm long. 8 m 61 cm is used. How much lace is left?

Step 1: Subtract: 19 m 82 cm - 8 m 61 cm. Step 2: cm: 82-61=21. m: 19-8=11. Answer: 11 m 21 cm lace left.

Q4. A journey is 950 km 460 m long. 900 km 230 m of it is by bus and the rest by taxi. How much of the journey is by taxi?

Step 1: Subtract: 950 km 460 m - 900 km 230 m. Step 2: m: 460-230=230. km: 950-900=50. Answer: 50 km 230 m by taxi.

Try it activity (book page 98): Two routes join two places. Route-1 is 5800 m long. Route-2 is 800 m shorter than Route-1, so Route-2 = 5800 - 800 = 5000 m. Both routes together = 5800 + 5000 = 10800 m.

Exercise 3 (book pages 102-103)

Fruit basket activity: Four baskets weigh 3 kg, 10 kg, 12 kg and 7 kg. The heaviest is basket 3 (12 kg). Total weight = 3 + 10 + 12 + 7 = 32 kg.

Q1. Add the masses: (i) 85 kg 245 g + 10 kg 134 g; (ii) 28 kg 325 g + 31 kg 550 g; (iii) 681 kg 845 g + 116 kg 102 g; (iv) 12 kg 340 g + 35 kg 257 g; (v) 962 kg 220 g + 36 kg 750 g; (vi) 342 kg 560 g + 37 kg 405 g.

Step 1: Add grams first, then kilograms. Answer: (i) 95 kg 379 g; (ii) 59 kg 875 g; (iii) 797 kg 947 g; (iv) 47 kg 597 g; (v) 998 kg 970 g; (vi) 379 kg 965 g.

Q2. Add: 10 kg 300 g + 12 kg 400 g.

Step 1: Add grams: 300+400=700. Step 2: Add kilograms: 10+12=22. Answer: 22 kg 700 g.

Q3. Add: 6 kg 250 g + 3 kg 500 g.

Step 1: Add grams: 250+500=750. Step 2: Add kilograms: 6+3=9. Answer: 9 kg 750 g.

Q4. Add: 15 kg 500 g + 12 kg 250 g.

Step 1: Add grams: 500+250=750. Step 2: Add kilograms: 15+12=27. Answer: 27 kg 750 g.

Exercise 4 (book page 106)

Q1. Subtract the masses: (i) 29 kg 750 g - 18 kg 250 g; (ii) 9 kg 763 g - 7 kg 250 g; (iii) 87 kg 986 g - 66 kg 350 g; (iv) 76 kg 565 g - 34 kg 324 g; (v) 97 kg 850 g - 53 kg 340 g; (vi) 82 kg 677 g - 75 kg 500 g.

Step 1: Subtract grams first, then kilograms, borrowing where needed. Answer: (i) 11 kg 500 g; (ii) 2 kg 513 g; (iii) 21 kg 636 g; (iv) 42 kg 241 g; (v) 44 kg 510 g; (vi) 7 kg 177 g.

Q2. A box has 27 kg 350 g of chocolates. 16 kg 250 g are sold. How many chocolates are left?

Step 1: Subtract: 27 kg 350 g - 16 kg 250 g. Step 2: g: 350-250=100. kg: 27-16=11. Answer: 11 kg 100 g chocolates left.

Q3. A shop has 240 kg of meat. 200 kg is sold. How much meat is left?

Step 1: Subtract: 240 - 200 = 40. Answer: 40 kg meat left.

Exercise 5 (book pages 110-111)

Q1. Add the capacities: (i) 15 l 675 ml + 32 l 312 ml; (ii) 8 l 350 ml + 9 l 245 ml; (iii) 42 l 651 ml + 21 l 248 ml; (iv) 35 l 459 ml + 63 l 510 ml; (v) 73 l 342 ml + 23 l 610 ml; (vi) 54 l 800 ml + 25 l 125 ml.

Step 1: Add millilitres first, then litres. Answer: (i) 47 l 987 ml; (ii) 17 l 595 ml; (iii) 63 l 899 ml; (iv) 98 l 969 ml; (v) 96 l 952 ml; (vi) 79 l 925 ml.

Q2. Add: 3 l 240 ml + 5 l 350 ml.

Step 1: Add millilitres: 240+350=590. Step 2: Add litres: 3+5=8. Answer: 8 l 590 ml.

Q3. A house uses 35 litres of canola oil and 15 litres of soybean oil. How much oil does the house use in all?

Step 1: Add: 35 + 15 = 50. Answer: 50 litres.

Q4. Add: 3 l 500 ml + 4 l.

Step 1: Add litres: 3+4=7, millilitres stay 500. Answer: 7 l 500 ml.

Exercise 6 (book pages 113-114)

Q1. Subtract the capacities: (i) 18 l 655 ml - 12 l 321 ml; (ii) 8 l 742 ml - 7 l 421 ml; (iii) 45 l 758 ml - 21 l 000 ml; (iv) 25 l 450 ml - 21 l 300 ml; (v) 75 l 915 ml - 61 l 800 ml; (vi) 58 l 334 ml - 35 l 621 ml.

Step 1: Subtract millilitres first, then litres, borrowing where needed. For (vi): 334 ml is less than 621 ml, so borrow 1 litre (1000 ml): 1334 - 621 = 713 ml, and litres 57 - 35 = 22. Answer: (i) 6 l 334 ml; (ii) 1 l 321 ml; (iii) 24 l 758 ml; (iv) 4 l 150 ml; (v) 14 l 115 ml; (vi) 22 l 713 ml.

Q2. A pot has 2000 ml of tea. 1500 ml is served. How much tea is left?

Step 1: Subtract: 2000 - 1500 = 500. Answer: 500 ml tea left.

Q3. A container has 2 l 450 ml. 1 l 200 ml is drunk. How much is left?

Step 1: Subtract: 2 l 450 ml - 1 l 200 ml. Step 2: ml: 450-200=250. l: 2-1=1. Answer: 1 l 250 ml left.

Q4. Subtract: 15 l 500 ml - 3 l.

Step 1: Subtract litres: 15-3=12, millilitres stay 500. Answer: 12 l 500 ml.

Review Exercise (book pages 115-117)

Q1 (i). What is the appropriate unit to determine the length of a needle?

(a) kilometre

(b) metre

✓ (c) centimetre

(d) millilitre

Q1 (ii). How many millilitre of water is saved after extracting 2000 ml of water from 3500 ml of water?

(a) 500 ml

(b) 1000 ml

✓ (c) 1500 ml

(d) 2000 ml

Q1 (iii). What is standard unit of mass?

(a) metre

(b) litre

(c) kilometre

✓ (d) kilogram

Q1 (iv). What is abbreviation of units of litre?

(a) ml

(b) g

✓ (c) l

(d) kg

Q1 (v). If I had two cans of 800 ml juice, what would be the total quantity?

(a) 200 ml

(b) 1000 ml

(c) 800 ml

✓ (d) 1600 ml

Q2. Solve: (i) 5 m 35 cm + 3 m 42 cm; (ii) 7 km 219 cm + 3 km 340 cm (as printed in the book, which mixes km with cm); (iii) 8 m 42 cm - 3 m 32 cm; (iv) 9 km 695 m - 5 km 362 m.

Step 1: Add or subtract the smaller unit first, then the larger unit. Answer: (i) 8 m 77 cm; (ii) 10 km 559 cm (as printed); (iii) 5 m 10 cm; (iv) 4 km 333 m.

Q3. Solve: (i) 4 kg 490 g + 3 kg 507 g; (ii) 7 kg 600 g + 6 kg 250 g; (iii) 9 kg 500 g - 7 kg 300 g; (iv) 15 kg 750 g - 11 kg 250 g.

Answer: (i) 7 kg 997 g; (ii) 13 kg 850 g; (iii) 2 kg 200 g; (iv) 4 kg 500 g.

Q4. Solve: (i) 8 l 780 ml + 5 l 217 ml; (ii) 7 l 500 ml + 4 l 250 ml; (iii) 9 l 300 ml - 4 l 200 ml; (iv) 6 l 500 ml - 5 l 200 ml.

Answer: (i) 13 l 997 ml; (ii) 11 l 750 ml; (iii) 5 l 100 ml; (iv) 1 l 300 ml.

Q5. Add: 100 kg + 80 kg (rice).

Step 1: Add: 100 + 80 = 180. Answer: 180 kg rice.

Q6. A ribbon is 350 metre long. 120 metre is cut off. How much ribbon is left?

Step 1: Subtract: 350 - 120 = 230. Answer: 230 metre ribbon left.

Q7. 5 litres are needed but only 4 litres are present. How much more is needed?

Step 1: Subtract: 5 - 4 = 1. Answer: 1 litre more needed.

Unit 5: Measurement - Time (book pages 118-132)

Exercise 1 (book pages 122-123)

Q1. For each clock picture, write the time and whether it is a.m. or p.m.: (i) morning 8:30; (ii) evening 4:40; (iii) school morning 7:25; (iv) evening 7:25; (v) dinner 8:35.

Step 1: Read the hour hand (short) and the minute hand (long). Step 2: Morning times are a.m.; afternoon and evening times are p.m. Answer: (i) 8:30 a.m. (morning). (ii) 4:40 p.m. (evening). (iii) 7:25 a.m. (school morning). (iv) 7:25 p.m. (evening, digital 07:25). (v) 8:35 p.m. (dinner, digital 08:35).

Q2. Read the time on each analogue clock: nine clocks as shown in the book.

Step 1: The short hand shows the hour; the long hand shows the minutes (each mark = 1 minute, each number = 5 minutes). Answer: (i) 5:00; (ii) 6:40; (iii) 3:00; (iv) 4:05; (v) 5:40; (vi) 10:50; (vii) 8:55; (viii) 2:45; (ix) 1:30.

Exercise 2 (book page 126)

Use the calendar in the book (January 1 is a Friday) to answer: (i) What day is January 31? (ii) What is the date of the second Wednesday of April? (iii) What day is 3rd December? (iv) What is the date of the last Friday of February? (v) What day is 23rd March?

Step 1: Find the month in the calendar. Step 2: Count the weeks and days.

Answer: (i) January 31 = Sunday. (ii) Second Wednesday of April = 14 April (April 1 is a Thursday). (iii) 3rd December = Friday (December 1 is a Wednesday). (iv) Last Friday of February = 26 February. (v) 23rd March = Tuesday (March 1 is a Monday).

Exercise 3 (book page 128)

Q1. Add the hours: (i) 5 + 3; (ii) 6 + 4; (iii) 10 + 5; (iv) 10 + 8; (v) 7 + 8; (vi) 15 + 5.

Answer: (i) 8 hours; (ii) 10 hours; (iii) 15 hours; (iv) 18 hours; (v) 15 hours; (vi) 20 hours.

Q2. Add: 5 hours + 2 hours.

Step 1: Add: 5 + 2 = 7. Answer: 7 hours.

Q3. Add: 10 hours + 8 hours.

Step 1: Add: 10 + 8 = 18. Answer: 18 hours.

Q4. Add: 9 hours + 8 hours.

Step 1: Add: 9 + 8 = 17. Answer: 17 hours.

Exercise 4 (book page 130)

Q1. Subtract the hours: (i) 8 - 5; (ii) 18 - 7; (iii) 15 - 6; (iv) 18 - 11; (v) 16 - 10; (vi) 21 - 8.

Answer: (i) 3 hours; (ii) 11 hours; (iii) 9 hours; (iv) 7 hours; (v) 6 hours; (vi) 13 hours.

Q2. (a) How many more hours is 4 hours than 2 hours? (b) Areesha starts her work at 11:00 a.m. and works for 2 hours. When does she stop?

Step 1: Subtract: 4 - 2 = 2 more hours. Step 2: Add the working time to the starting time: 11:00 a.m. + 2 hours = 1:00 p.m. Answer: 2 more hours; she stops at 1:00 p.m.

Q3. Subtract: 8 hours - 5 hours.

Step 1: Subtract: 8 - 5 = 3. Answer: 3 more hours.

Q4. 8 hours and 5 hours: how many hours less is the smaller time? (The book's question asks it the other way round: the time for Mathematics is less.)

Step 1: Subtract: 8 - 5 = 3. Answer: 3 hours less for Mathematics.

Review Exercise (book pages 131-132)

Q1. Match each analogue clock with its digital time: 2:15, 7:55, 4:29, 11:20.

Step 1: Read the hour and minute hands of each clock. Step 2: The third clock reads 4:29 (hour hand between 4 and 5, minute hand just before 6). Answer: 2:15 = 02:15; 7:55 = 07:55; 4:29 = 04:29; 11:20 = 11:20.

Q2. Add: 3 hours + 2 hours + 1 hour.

Step 1: Add: 3 + 2 + 1 = 6. Answer: 6 hours.

Q3. Use the calendar: (i) First Monday of May; (ii) Day of 15th April; (iii) Second Friday of August; (iv) Third Saturday of November; (v) Day of 31st December.

Step 1: Find the month and count the weeks. Answer: (i) First Monday of May = 3 May (May 1 is a Saturday). (ii) 15th April = Thursday. (iii) Second Friday of August = 13 August (August 1 is a Sunday). (iv) Third Saturday of November = 20 November (November 1 is a Monday). (v) 31st December = Friday (December 1 is a Wednesday).

Q4. Add: 5 hours + 6 hours.

Step 1: Add: 5 + 6 = 11. Answer: 11 hours.

Q5. Subtract: 13 hours - 6 hours.

Step 1: Subtract: 13 - 6 = 7. Answer: 7 hours.

Unit 6: Geometry (book pages 133-157)

Exercise 1 (book page 139)

Q1. Identify each figure: AB (dots at both ends); CD (dot at C, arrow at D); X (a single dot); EF (arrows at both ends); YX (arrow at Y, dot at X); PQ (dots at both ends).

Step 1: A figure with two end points is a line segment. Step 2: A figure with one end point and one arrow is a ray. Step 3: A figure with arrows at both ends is a line. Step 4: A single dot is a point. Answer: AB = line segment; CD = ray; X = point; EF = line; YX = ray; PQ = line segment.

Q2. Measure each line segment with a ruler: AB, CD, EF. (Measured from the PDF page scale; on a differently printed page the readings may differ slightly.)

Step 1: Place the ruler's 0 mark at one end of the segment. Step 2: Read the mark at the other end in centimetres and millimetres. Answer: AB = 8.3 cm (83 mm); CD = 6.7 cm (67 mm); EF = 10.2 cm (102 mm).

Q3. Draw line segments of: (i) 1.9 cm; (ii) 4.2 cm; (iii) 5.6 cm.

Step 1: Mark the starting point at 0 on the ruler. Step 2: Mark the end point at the given length. Step 3: Join the two points with a straight line and label the ends. Answer: (i) A segment 1.9 cm long. (ii) A segment 4.2 cm long. (iii) A segment 5.6 cm long.

Exercise 2 (book pages 147-149)

Q1. Identify each figure: AB (dots at both ends); Y (a single dot); EF (dot at one end, arrow at the other); CD (dots at both ends); PQ (arrows at both ends); RS (dot at one end, arrow at the other); X (a single dot); MN (arrows at both ends); JK (dots at both ends).

Step 1: Two end points means line segment; one end point and one arrow means ray; arrows at both ends means line; a single dot means point. Answer: AB = line segment; Y = point; EF = ray; CD = line segment; PQ = line; RS = ray; X = point; MN = line (arrows at both ends, not a ray); JK = line segment.

Q2. For each closed figure, write the length of each side and its perimeter: (i) Square ABCD; (ii) Rectangle PQRS; (iii) Triangle XYZ. (Side lengths measured from the PDF page scale; on a differently printed page the readings may differ slightly.)

Step 1: Measure each side. Step 2: Perimeter = sum of all sides. Answer: (i) Square ABCD: AB = BC = CD = AD = 4 cm. Perimeter = 4 + 4 + 4 + 4 = 16 cm. (ii) Rectangle PQRS: PQ = 6.2 cm, QR = 3.5 cm, RS = 6.2 cm, PS = 3.5 cm. Perimeter = 6.2 + 3.5 + 6.2 + 3.5 = 19.4 cm. (iii) Triangle XYZ: XY = 4.6 cm, YZ = 4.6 cm, XZ = 4.6 cm. Perimeter = 4.6 + 4.6 + 4.6 = 13.8 cm.

Q3. Find the perimeter of: (i) a square of side 3 cm; (ii) a square of side 4 cm.

Step 1: Perimeter of a square = 4 x side. Answer: (i) 4 x 3 = 12 cm. (ii) 4 x 4 = 16 cm.

Q4. Find the perimeter of the rectangles: (i) 4 cm by 5 cm; (ii) 4 cm by 6 cm.

Step 1: Perimeter of a rectangle = 2 x (length + width). Answer: (i) 2 x (4 + 5) = 2 x 9 = 18 cm. (ii) 2 x (4 + 6) = 2 x 10 = 20 cm.

Q5. Find the perimeter of the triangles: (i) sides 4 cm, 5 cm, 3 cm; (ii) sides 4 cm, 4 cm, 5 cm.

Step 1: Perimeter of a triangle = sum of its three sides. Answer: (i) 4 + 5 + 3 = 12 cm. (ii) 4 + 4 + 5 = 13 cm.

Q6. In the circle figures, identify: (i) dot A at the centre; (ii) the line from O to A (centre to circumference); (iii) the line from B to C (circumference to centre); (iv) the line from C to D passing through O (end to end through the centre).

Step 1: The centre point is the centre. Step 2: Centre to circumference is the radius. Step 3: End to end through the centre is the diameter. Answer: (i) Centre; (ii) Radius; (iii) Radius; (iv) Diameter.

Exercise 3 (book page 152)

Q1. How many lines of symmetry does each figure have? (i) half-rounded rectangle; (ii) square; (iii) letter A; (iv) tall isosceles triangle; (v) elongated hexagon; (vi) kite.

Step 1: A line of symmetry divides a figure into two exactly matching halves. Step 2: Fold mentally along each possible line. Answer: (i) 1 horizontal line. (ii) 4 lines. (iii) 1 vertical line. (iv) 1 vertical line (the triangle is taller than it is wide, so it is not equilateral). (v) 2 lines (horizontal and vertical). (vi) 1 vertical line.

Q2. How many lines of symmetry are shown by the dashed lines? (i) circle; (ii) rhombus; (iii) flower; (iv) plus-shape; (v) letter H; (vi) pentagon.

Step 1: Count the dashed lines drawn on each figure. Answer: (i) circle: 2; (ii) rhombus: 2; (iii) flower: 1; (iv) plus-shape: 4; (v) H: 2; (vi) pentagon: 5.

Exercise 4 (book page 155)

Complete the table of faces, edges and vertices: Cube (yellow); Cuboid (blue); Square pyramid (purple); Dice (red); Tissue box; Triangular prism (blue wedge).

Step 1: Count the flat faces. Step 2: Count the edges where two faces meet. Step 3: Count the corners (vertices). Answer: Cube: 6 faces, 12 edges, 8 vertices. Cuboid: 6 faces, 12 edges, 8 vertices. Square pyramid: 5 faces, 8 edges, 5 vertices. Dice: Cube, 6 faces, 12 edges, 8 vertices. Tissue box: Cuboid, 6 faces, 12 edges, 8 vertices. Triangular prism: 5 faces, 9 edges, 6 vertices.

Review Exercise (book pages 156-157)

Q1 (i). Number of sides in a quadrilateral are:

(a) 1

(b) 2

(c) 3

✓ (d) 4

Q1 (ii). In a cube, number of edges are:

(a) 2

(b) 6

(c) 8

✓ (d) 12

Q1 (iii). A triangle has _____ vertices:

(a) 2

✓ (b) 3

(c) 4

(d) 5

Q1 (iv). Given figure AB (dots at both ends) is:

(a) Point

(b) line

✓ (c) line segment

(d) Ray

Q1 (v). Given figure AB (arrows at both ends) is:

✓ (a) line

(b) Ray

(c) line segment

(d) Point

Line of symmetry divides any shape into two equal parts.

Line segment has two end points.

A square shape has 4 sides and 4 vertices.

The sum of all sides of any closed shape is perimeter.

Perimeter of a rectangle = 2 x (length + width).

Q3. Draw the line segments according to the given measurements: (i) RS = 7 cm; (ii) XY = 3 cm; (iii) AB = 5 cm.

Step 1: Mark the starting point at 0 on the ruler. Step 2: Mark the end point at the given length and join. Answer: (i) RS = 7 cm. (ii) XY = 3 cm. (iii) AB = 5 cm.

Q4. Ahmad walks a square shaped ground with length 249 m. How much distance does he cover in one round?

Step 1: One round of a square = its perimeter = 4 x side. Step 2: Multiply: 4 x 249 = 996. Answer: 996 m.

Q5. Classroom door is 210 cm long and 120 cm wide. Find its perimeter.

Step 1: Perimeter of a rectangle = 2 x (length + width). Step 2: Add: 210 + 120 = 330. Step 3: Multiply: 2 x 330 = 660. Answer: 660 cm.

Q6. Find the perimeter of a triangular field with lengths of 15 m, 25 m and 40 m.

Step 1: Add the three sides: 15 + 25 + 40 = 80. Answer: 80 m.

Unit 7: Data Handling (book pages 158-171)

Teaching-page question panels (book pages 162-163, 166)

Carroll panel (book page 162): Sort the numbers into Even/ Odd and Divisible by 5 / Not divisible by 5. (i) Smallest even number divisible by 5 (given in the book); (ii) odd numbers divisible by 5; (iii) smallest even number not divisible by 5; (iv) greatest odd number not divisible by 5.

Step 1: Check each number for even/odd and for divisibility by 5. Answer: Even and divisible by 5: 10, 20, 30. Odd and divisible by 5: 5, 15, 25. Even and not divisible by 5: 4, 8, 14. Odd and not divisible by 5: 3, 9, 19, 21. (i) Smallest even divisible by 5 = 10 (given). (ii) Odd divisible by 5: 5, 15, 25. (iii) Smallest even not divisible by 5 = 4. (iv) Greatest odd not divisible by 5 = 21.

Tally panel (book pages 162-163): Animals seen: Monkey 7, Lion 6, Bear 4, Zebra 6, Elephant 2. (i) Least (given in the book); (ii) greatest; (iii) equal; (iv) monkey + lion; (v) total.

Step 1: Read each tally count. Answer: (i) Least = Elephant (given). (ii) Greatest = Monkey (7). (iii) Equal = Lion and Zebra (6 each). (iv) Monkey + Lion = 7 + 6 = 13. (v) Total = 7 + 6 + 4 + 6 + 2 = 25.

Sports pictograph panel (book page 166; 1 picture = 2 students): Hockey 5 pictures, Football 6 pictures, Cricket 4 pictures, Tennis 3 pictures, Table Tennis 2 pictures. (i) hockey; (ii) cricket; (iii) most; (iv) least; (v) tennis.

Step 1: Multiply the number of pictures by 2. Answer: Hockey = 10; Football = 12; Cricket = 8; Tennis = 6; Table Tennis = 4. (i) Hockey = 10. (ii) Cricket = 8. (iii) Most = Football (12). (iv) Least = Table Tennis (4). (v) Tennis = 6.

Exercise 1 (book pages 163-164)

Q1. Sort the pictured fruits and vegetables in a Carroll diagram. (Tomato is placed with the vegetables, following the usual school convention; botanically it is a fruit.)

Step 1: Identify each pictured item. Step 2: Put fruits in one group and vegetables in the other. Answer: Fruits: mango, plum, peach, pear, cherry, watermelon, apple (7). Vegetables: bitter gourd, capsicum, eggplant, carrot, green chilli, tomato (6).

Q2. Sort (10, 18, 22, 25, 29, 30, 35, 37, 45, 43, 48, 52) into a Carroll diagram: Even/Odd and Divisible by 5 / Not divisible by 5.

Step 1: Check each number for even/odd. Step 2: Check divisibility by 5 (ends in 0 or 5). Answer: Even and divisible by 5: 10, 30. Odd and divisible by 5: 25, 35, 45. Even and not divisible by 5: 18, 22, 48, 52. Odd and not divisible by 5: 29, 37, 43.

Q3. Sort into a Carroll diagram (less than 25 / greater than 25 and divisible by 7 / not divisible by 7): (i) greater and divisible by 7; (ii) greater and not divisible by 7; (iii) less and divisible by 7; (iv) less and not divisible by 7.

Step 1: Compare each number with 25. Step 2: Check divisibility by 7. Answer: (i) Greater and divisible by 7: 28, 35, 42. (ii) Greater and not divisible by 7: 27, 29, 38, 43. (iii) Less and divisible by 7: 7, 14, 21. (iv) Less and not divisible by 7: 5, 9, 15, 19.

Q4. A die is rolled 20 times: 1, 3, 5, 6, 3, 2, 4, 5, 3, 2, 4, 6, 3, 4, 3, 4, 2, 5, 1, 6. Make a tally chart for each number.

Step 1: Count how many times each number from 1 to 6 appears. Step 2: Check the total is 20. Answer: 1: 2 times; 2: 3 times; 3: 5 times; 4: 4 times; 5: 3 times; 6: 3 times. Total = 20.

Q5. Birthday tally: January 3, February 6, March 10, April 7, May 4. (i) Least; (ii) greatest; (iii) January + April; (iv) total.

Step 1: Compare the counts. Answer: (i) Least = January (3). (ii) Greatest = March (10). (iii) January + April = 3 + 7 = 10. (iv) Total = 3 + 6 + 10 + 7 + 4 = 30.

Exercise 2 (book pages 167-168)

Q1. Pictograph of students present (1 picture = 1 student): Monday 4, Tuesday 3, Wednesday 5, Thursday 6, Friday 2. (i) Monday; (ii) Tuesday; (iii) most; (iv) least; (v) Wednesday + Thursday.

Step 1: Count the pictures for each day. Answer: (i) Monday = 4. (ii) Tuesday = 3. (iii) Most = Thursday (6). (iv) Least = Friday (2). (v) Wednesday + Thursday = 5 + 6 = 11.

Q2. Pictograph of cars sold (1 picture = 100 cars): 2008: 4 pictures, 2009: 5 pictures, 2010: 3 pictures, 2011: 4 pictures, 2012: 2 pictures. (i) 2008; (ii) 2010; (iii) most; (iv) least; (v) equal years.

Step 1: Multiply the number of pictures by 100. Answer: 2008 = 400; 2009 = 500; 2010 = 300; 2011 = 400; 2012 = 200. (i) 2008 = 400. (ii) 2010 = 300. (iii) Most = 2009 (500). (iv) Least = 2012 (200). (v) Equal = 2008 and 2011 (400 each).

Review Exercise (book pages 169-170)

Q1. Sort (12, 15, 16, 18, 17, 19, 21, 23, 28, 30, 32, 37, 39) into a Carroll diagram: Less than 20 / Greater than 20 and Even / Odd.

Step 1: Compare each number with 20. Step 2: Check even/odd. Answer: Less than 20 and Even: 12, 16, 18. Less than 20 and Odd: 15, 17, 19. Greater than 20 and Even: 28, 30, 32. Greater than 20 and Odd: 21, 23, 37, 39.

Q2. Tally of favourite subjects: Urdu 7, Science 8, English 10, Mathematics 6, General Knowledge 4. (i) least; (ii) most; (iii) Science; (iv) Urdu; (v) Mathematics.

Step 1: Compare the counts. Answer: (i) Least = General Knowledge (4). (ii) Most = English (10). (iii) Science = 8. (iv) Urdu = 7. (v) Mathematics = 6.

Q3. Pictograph of animals (1 picture = 2 animals): Deer 4 pictures, Elephant 3 pictures, Monkey 4 pictures, Cheetah 2 pictures. (i) greatest; (ii) least; (iii) elephants + monkeys; (iv) total.

Step 1: Multiply the number of pictures by 2. Answer: Deer = 8; Elephant = 6; Monkey = 8; Cheetah = 4. (i) Greatest = Deer and Monkey (8 each, tie). (ii) Least = Cheetah (4). (iii) Elephants + monkeys = 6 + 8 = 14. (iv) Total = 8 + 6 + 8 + 4 = 26.